Selecting streams with the -map option

The -map option is used to choose which streams from the input(s) should be included in the output(s). The `-map` option can also be used to exclude specific streams with *negative mapping*.

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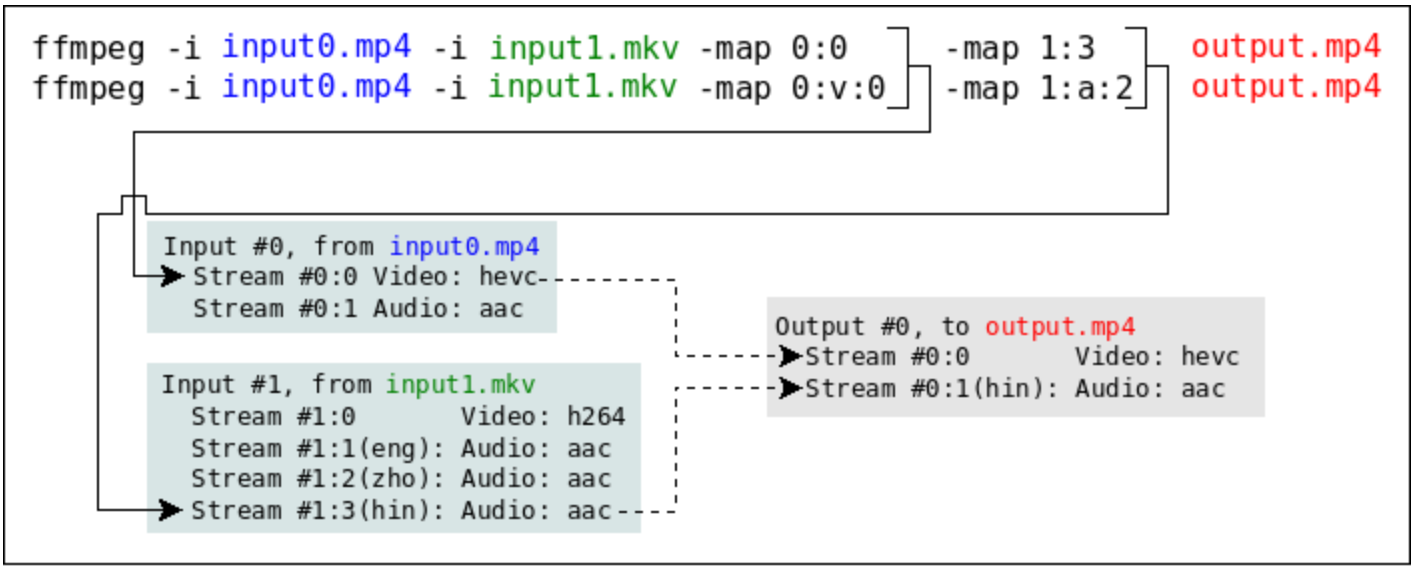
See also



The commands in the diagram above will select the video from `input0.mp4` and the 3rd audio stream from `input1.mkv` to `output.mp4`. The commands do the same thing, but use different syntax to map streams:

- The top command does not use any *stream specifiers*. This is an absolute method of mapping and will choose a specific stream regardless of type. `-map 1:3` refers to "*second input:fourth stream*".
- The bottom command includes stream specifiers (`v` for video, `a` for audio) to limit the streams to a specific type. `-map 1:a:2` refers to "*second input:audio only:third audio stream*".

Simple examples

- `-map 0` From input index #0 (the 1st input) select all streams.
- `-map 1:a` From input index #1 (the 2nd input) select all audio streams.
- `-map 3:s:4` From input index #3 (the 4th input) select subtitle stream index #4 (the fifth subtitle stream).
- `-map 0 -map -0:s` Will select all streams from input index #0 (the 1st input) except subtitles. The `-` indicates *negative mapping*.

Default behavior

If you do not use the `-map` option then the default stream selection behavior will automatically choose streams.

- Default stream selection will **not** automatically choose **all** of the streams.
- Only 1 stream per type will be selected. For example, if the input has 3 video streams it will only choose 1.
- The default stream selection will choose streams based upon specific criteria.

Using the `-map` option disables the default stream selection behavior and allows you to manually choose streams.

Syntax

The syntax is:

```
-map input_file_index:stream_type_specifier:stream_index
```

- `input_file_index` refers to an input and by default will include all of its streams.
- `stream_type_specifier` *(optional)* is a stream specifier which will limit the selection to a specific stream type. Stream specifiers include `v`ideo, `a`udio, `s`ubtitles, `d`ata, a`t`tachments. Most of the examples below use stream specifiers.
- `stream_index` *(optional)* will select a specific stream. If `stream_type_specifier` is used then the `stream_index` only refers to the selected stream type and ignores all other stream types.

> **Note:** input_file_index numbers start from 0. The first -i input or stream is #0, second -i input or stream is #1, third -i input or stream is #2, etc.

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Modifiers

- A `-` character before the `input_file_index`, such as `-map -0:v`, creates a *negative mapping*. This excludes streams already selected by a previous map. See example.
- A trailing `?`, such as `-map 1:a?`, will allow the map to be optional. If the map matches no streams the map will be ignored instead of failing. Note the mapping will still fail if an invalid `input_file_index` is used; such as if the map refers to an input that does not exist. See example.

Order

The order of `-map` options is important:

- `-map` order determines the stream order in the output(s).
- Mapping is applied in order. This is useful when using negative mapping.

Examples

> **Tip:** You can add `-c copy` to many of these examples to enable stream copy mode. This is useful if you want to mux only and avoid encoding.

Choose all streams

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Single input:

```
ffmpeg -i input.avi -map 0 output.mkv
```

Multiple inputs:

```
ffmpeg -i input0.mkv -i input1.mp4 -i input2.wav -map 0 -map 1 -map 2 output.mkv
```

Using stream copy to remux (no encoding):

```
ffmpeg -i input.avi -map 0 -c copy output.mkv
```

Video streams only

```
ffmpeg -i input.mp4 -map 0:v output.mp4
```

Audio streams only

```
ffmpeg -i input.mp4 -map 0:a output.mp4
```

A specific video stream only

Choose the 5th video stream:

```
ffmpeg -i input.mkv -map 0:v:4 output.mp4
```

Note that counting starts from 0.

Video and audio from different files

```
ffmpeg -i video.mp4 -i audio.m4a -map 0:v -map 1:a output.mp4
```

Everything except audio

Using negative mapping:

```
ffmpeg -i input.mp4 -map 0 -map -0:a output.mp4
```

Specific language

Include all video and all streams with Spanish language metadata:

```
ffmpeg -i input.mkv -map 0:v -map 0:m:language:spa output.mp4
```

See List of ISO 639-1 codes for the 3 letter language names.

Choose outputs from filters

In this example the filtered video is named `[v]` and the filtered audio is named `[a]`:

```
ffmpeg -i input.mp4 -i logo.png -i audio.mp3 -filter_complex "[0]scale=1280:-1[b
```

You can use almost any arbitrary names to label the filter outputs.

Multiple outputs

Output video to `video.mp4`. Output audio stream index #1 (the 2nd audio stream) to `audio_1.mp3`. Output audio stream index #3 (the 4th audio stream) to `audio_3.wav`:

```
ffmpeg -i input.mkv -map 0:v video.mp4 -map 0:a:1 audio_1.mp3 -map 0:a:3 audio_3
```

Note that counting starts from 0.

Optional mapping

Adding a trailing question mark (`?`) to `-map` will ignore the mapping if the stream does not exist.

To map the video and audio streams and ignore the audio mapping if no audio streams exist:

```
ffmpeg -i input.webm -map 0:v -map 0:a? output.mkv
```

If `-map 0:a` is used instead, and there are no audio streams, then `ffmpeg` will fail with `Stream map '0:a' matches no streams. To ignore this, add a trailing '?' to the map.`.

Re-order streams

The order of your `-map` options determines the order of the streams in the output. In this example the input file has audio as stream #0 and video as stream #1 (which is possible but unusual). Example to re-position video so it is listed first, followed by the audio:

```
ffmpeg -i input.mp4 -map 0:v -map 0:a -c copy output.mp4
```

This example stream copies (re-mux) with `-c copy` to avoid re-encoding.

See also

- -map option documentation

▶ **Attachments** (1)

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